BEGINNER GUIDE

AETHER: how the website works

Camera flight on scroll, a cursor x-ray, depth photos, a lap and a configurator — and the plan to make the car HD with Blender.

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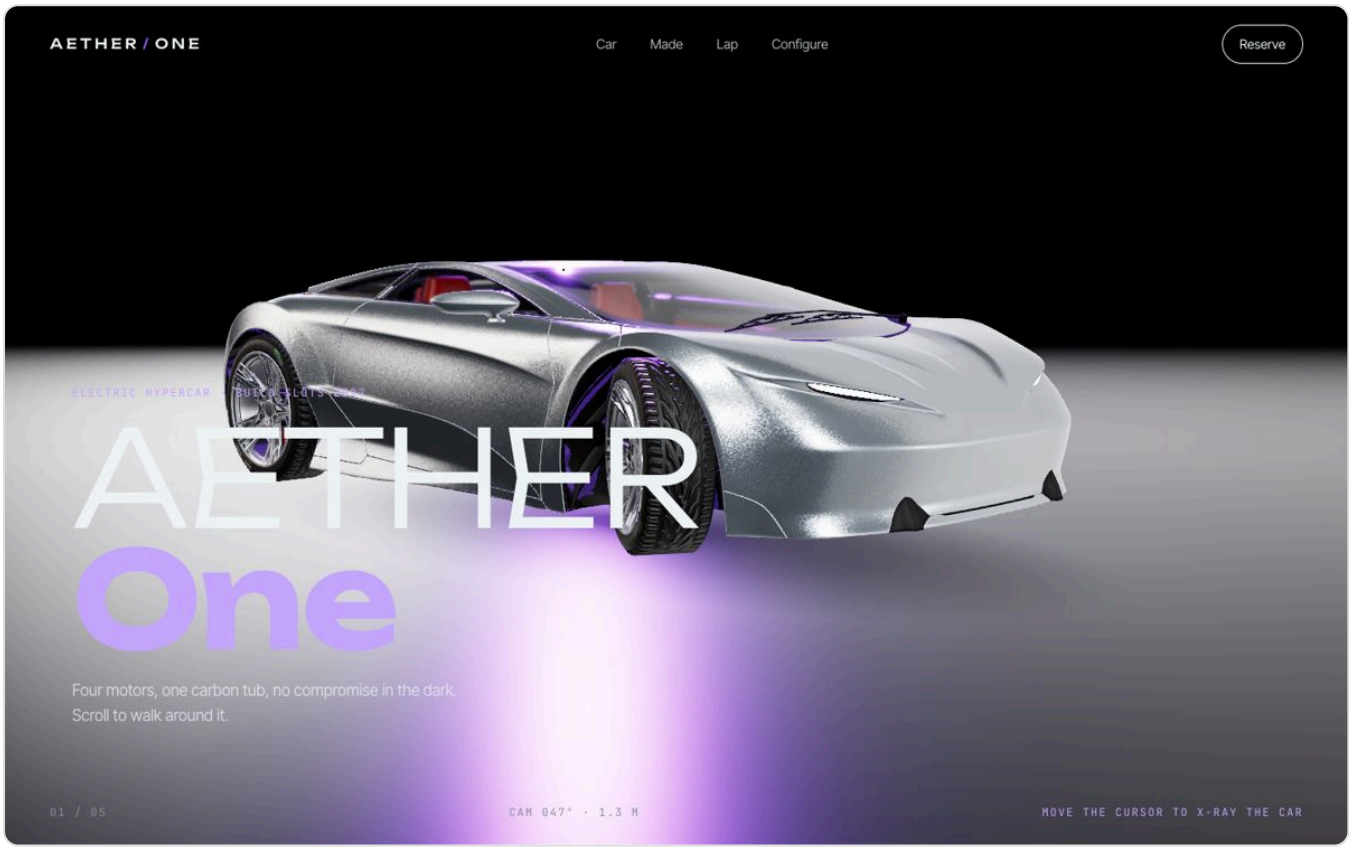
Written for Amartya Ghosh

1 Nov 2026 · aether-hypercar-3d.vercel.app

01 What you are looking at

AETHER One is a concept site for an imaginary electric hypercar. It is built **without React** — just HTML, CSS and plain JavaScript, plus three.js for 3D and Lenis for smooth scrolling. Vite runs it locally.

The car is a **real-time 3D model** (Khronos “Car Concept”, free with credit). That's also its weakness: free models look less real than film renders. The next version will use **pre-rendered HD frames from Blender** (see the last section).



The drive: the car in a dark studio, lit by a real studio photo (HDRI).

Part	What it does	File
The drive (600vh, pinned)	Camera flies around the car on a path; 5 chapters of text	src/hero.js
Cursor x-ray	Your cursor reveals the car's structure	src/hero.js
Made (sticky stack)	Four photo cards that shrink as the next covers them; photos raised into 3D	src/depthphoto.js + main.js
The lap	A race track drawn by scroll that brakes into corners	src/lap.js
Configure	Paint, wheels, packs, price, reserve form	src/config.js + main.js
The car itself	Loads the model, swaps logos, sets paint	src/car.js

02 The camera flight

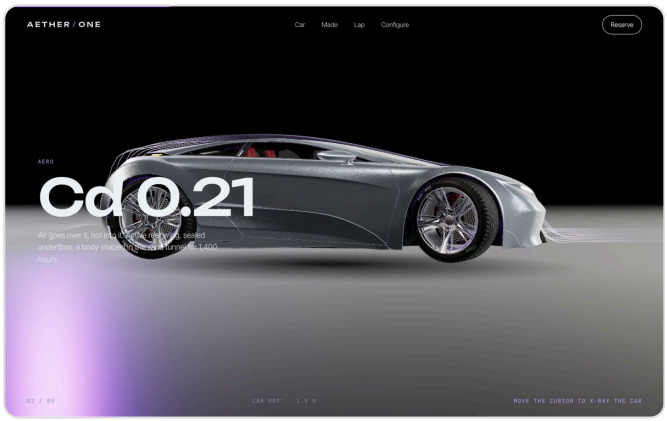
The section is 6 screens tall; its inside is sticky. Scroll progress `p` (0 → 1) picks a point on a **smooth curve through five camera positions** (front ¾, side, rear, top, close front). The camera then *eases* toward it each frame:

```
// src/hero.js
const KEYS = [
  { p: [6.6, 1.5, 6.1],  l: [-0.75, 0.55, 0.75] },  // camera position p, looking at l
  { p: [0.2, 0.95, 8.8], l: [-1.5, 0.6, 0] },
  …
];
const path = new THREE.CatmullRomCurve3(KEYS.map(k => new THREE.Vector3(...k.p)));
path.getPoint(p, target);                    // where the camera should be now
camPos.lerp(target, 1 - Math.exp(-dt * 6));  // glide there smoothly
camera.lookAt(camLook);
```

> Why `1 - exp(-dt·6)` instead of a fixed 0.1? It glides the same speed on a fast or slow computer.

03 The x-ray effect

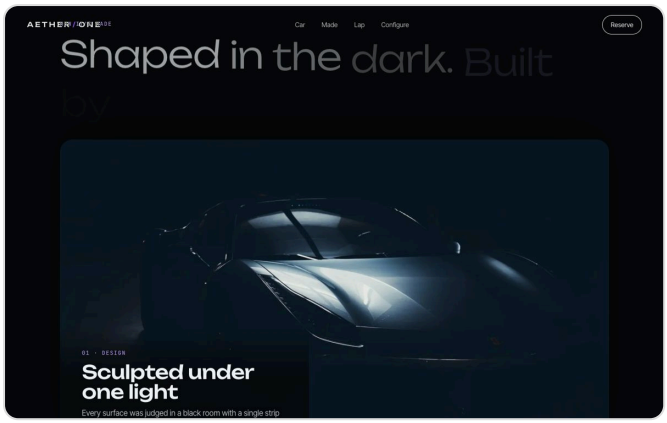


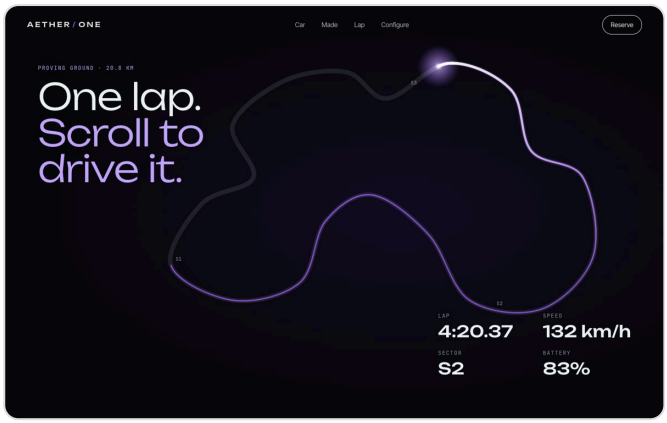


Left: the cursor x-rays the car. Right: aero lines that follow the real body shape.

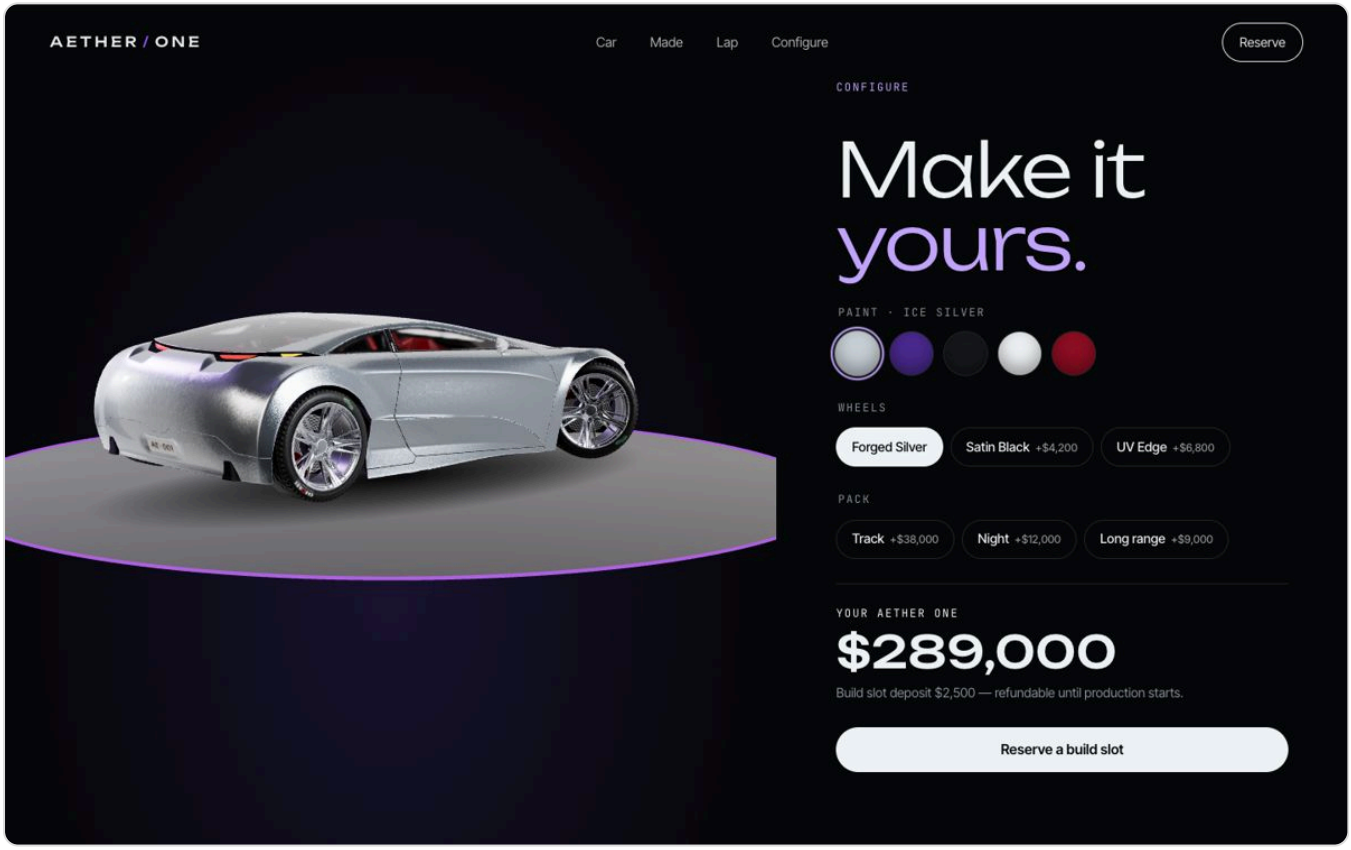
- Each frame the scene is drawn **twice** into hidden images: once normally, once with every surface turned into a glowing violet outline.
- A small invisible canvas follows your cursor and paints soft white blobs that fade over time — this is the **mask**.
- A final shader mixes the two images: where the mask is white you see the x-ray, elsewhere the paint, with a violet edge between.
- The aero lines were made by shooting rays straight down onto the car and following the heights they hit.

04 Sticky stack, lap and configurator





Made: cards recede as the next covers them; each photo is raised into 3D with a depth map. Lap: time follows the scroll, so the line slows in corners.



Configure: paint and wheels change the shared materials, so the hero car changes too.

- **Stack recede:** each card is `position: sticky`. When the next card's top comes up, the current card shrinks to 90% and darkens — computed from the next card's position.
- **Depth photos:** an AI model (Depth-Anything) made a grey depth image for each photo (white = near). The photo is laid on a dense mesh and every point is pushed toward you by its depth; the camera drifts with your mouse.
- **Lap:** the track is a smooth curve; tighter corners count as “slower”, so scrolling spends longer there — it looks like braking.

05 Next: HD car from Blender (the plan)

To make the car look like a film, we'll replace the live model in the hero with **~120 pre-rendered frames** from Blender:

- You download a high-quality car model into `~/Downloads/assets/car/`.
- A Python script runs Blender without opening it: imports the car, builds a dark studio with a real HDRI light and soft floor shadow, circles a camera around it, renders 1920×1080 frames with Cycles.
- The frames become WebP images; the site draws frame number `round(p × 119)` on a canvas as you scroll.
- The x-ray can stay: render a second set of frames with a wireframe/ghost material and mix them with the same cursor mask.

```
cd ~/Downloads/aether
npm install          # first time only
npm run dev          # opens on http://localhost:3007

# Publish a change
git add -A && git commit -m "describe the change" && git push
npx vercel --prod    # redeploy (or connect the repo in Vercel for automatic deploys)
# live: https://aether-hypercar-3d.vercel.app
```

06 Words you'll meet

Word	What it means here
Section / pinned section	A block of the page. *Pinned* (or *sticky*) means it stays on screen while you keep scrolling, so things inside it can animate.
Scroll progress (p)	A number from 0 to 1: how far you are through a section. Almost every animation is just “turn p into a position, size or opacity”.
Component	A reusable piece of a React page, written in a `.tsx` file (e.g. `Hero`, `Workstation`).
State	A value a component remembers (e.g. which tab is open). Changing it redraws that part of the page.
motion	The animation library: `motion.div` is a div that can animate; `useScroll` gives scroll progress; `useTransform` maps it to a value.
Lenis	Smooth scrolling: it makes the wheel feel soft and inertial instead of jumpy.
three.js / WebGL	Real 3D in the browser: a scene, a camera, lights and objects drawn by the graphics card every frame.
Shader (GLSL)	A tiny program that runs on the graphics card for every pixel (or every point) — used for special looks like night lights or glow.
Texture	An image wrapped onto a 3D object (e.g. the Earth's day map).
Glass / glassmorphism	A see-through panel: semi-transparent background + `backdrop-filter: blur()` + a thin bright border.

Word	What it means here
Tailwind classes	Short style words in `className`, e.g. `rounded-2xl` (round corners), `px-6` (side padding), `text-ink/60` (ink colour at 60%).